

Coffee Price Prediction: An Application of CNN-BLSTM Neural Networks

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Abstract- Coffee is one of the world's most popular beverages, and its production and demand have been steadily increasing in recent years. In 2020/21, worldwide coffee output hit 174.5 million bags, according to the International Coffee Organization. coffee year, which is a 1.9% increase from the previous year. The demand for coffee is driven by various factors, including changing consumer preferences, economic conditions, and demographic trends. In particular, the growing popularity of specialty coffee and the increasing consumption of coffee in emerging economies have contributed to the growth in demand. However, the coffee market has also faced challenges such as climate change, which can affect coffee production by altering the growing conditions, and the COVID-19 pandemic, which has disrupted supply chains and caused fluctuations in prices. In terms of regional When it comes to coffee output, Brazil leads the globe, followed by Vietnam, Colombia, and Indonesia. These countries collectively account for more than 60% of global coffee production. The United States, Germany, and Japan are the largest importers of coffee. Overall, coffee continues to be an important commodity in the global market, with a significant impact on the economies of producing countries and the daily routines of consumers around the world. In this article, we propose a fresh method of coffee price prediction using the The BLSTM (bidirectional long short-term memory) and CNN (convolutional neural networks) models. We start by collecting historical coffee price data from publicly available sources and preprocess it using feature engineering techniques. The The collected data was then split into training and validation sets and testing sets and feed it into the proposed CNN-BLSTM model. The CNN extraction by using layers the relevant features from the input data and reduce its dimensionality, while the BLSTM layers learn temporal dependencies in the data and

capture long-term patterns. The outputs from the BLSTM layers are then fed into fully connected layers, which output the final price prediction. We Use measures like MSE, RMSE, and MAE to measure how far off you are from your target assess how well our suggested model performs (MAE) in both the test and validation data. Our According to the obtained CNN-BLSTM model outperforms several other state-of-the-art machine learning models, including traditional time-series models, on the same dataset. Overall, our approach demonstrates the effectiveness of combining CNN and BLSTM models for coffee price prediction and can be extended to other related forecasting problems.

Keywords: Coffee price prediction, Convolutional Neural Network, BLSTM, etc.,

I. INTRODUCTION

Coffee enjoys widespread acclaim as a delicious and refreshing drink. global consumer base that spans across various age groups, cultures, and geographic regions. Coffee production is a major industry that supports millions of farmers, traders, and businesses around the world. The cultivation and production of coffee involves a complex supply chain that includes farmers, processors, exporters, importers, roasters, and retailers. Coffee is grown in more than 50 countries, with the majority of the world's production coming from Brazil, Vietnam, Colombia, Indonesia, and Ethiopia.

Now, the coffee market is a significant an estimated 25 million persons make a contribution to the global economy involved Coffee in a production commerce on a global scale. The industry generates billions of

dollars in revenue annually, making it a vital source of income for many developing countries. On the consumer side, coffee is enjoyed by people of all ages and backgrounds, with an estimated 2.25 billion cups of coffee drunk every day. In many cultures, coffee is an essential part of daily life, with social and cultural significance beyond its role as a beverage.

The popularity of coffee has led to the creation of a diverse range of coffee products and variations, from traditional brewed coffee to espresso, cappuccino, latte, and more. The demand for specialty coffee and high-quality beans has also increased in recent years, with consumers seeking out unique and distinctive flavors and aromas. Overall, the coffee industry is a vital part of global trade and commerce, with a wide-ranging impact on both producers and consumers around the world.

There are various types of coffee-based businesses, including coffee shops, roasteries, wholesalers, and online retailers. The turnover of these businesses can vary greatly depending on factors such as location, size, and target market. Coffee shops are one of the most common types of coffee-based businesses, and their turnover can range from a few thousand to several million dollars per year. The turnover of a coffee shop is dependent on factors such as the number of customers, the price point of products, and the location. In high-traffic areas with a large customer base, a coffee shop can generate significant turnover through the sale of coffee, tea, pastries, and other food items.

Roasteries, on the other hand, typically generate revenue through the sale of roasted coffee beans to wholesale customers, coffee shops, and other retailers. The turnover of a roastery can change based on things including how big the business, the quality of the beans, and the number of customers. A small roastery may have a turnover of a few hundred thousand dollars per year, while a larger operation may generate several million dollars in revenue. Wholesalers in the coffee industry typically sell roasted coffee beans, equipment, and other supplies to coffee shops, restaurants, and other businesses. The turnover of a coffee wholesaler size-dependent variations in operation and the total amount of customers. A small wholesaler may have a turnover of a few hundred thousand dollars per year, while a larger wholesaler may generate several million dollars in revenue.

Online retailers are becoming increasingly popular in the coffee industry, with many consumers opting to purchase coffee and coffee-related products online. The turnover of an online coffee retailer can change based on things including how big the operation, the number of customers, and the variety of products offered. A small online retailer may have a turnover of a few hundred thousand dollars per year, while a larger operation may generate several million dollars in revenue.

Coffee price prediction is an essential subject for study for both coffee producers and traders. The deployment of advanced Machine learning (ML) and artificial intelligence (AI) are two examples of computational approaches. Both BLSTM (bidirectional long short-term memory) and convolutional (deep) neural networks are used (CNN), has shown promise in accurately predicting coffee prices.

CNN-BLSTM neural networks combine the strengths of CNNs, which are effective in processing spatial features, and BLSTM, which is effective in processing time-series data. The combination of these two models allows for the extraction of relevant features from both spatial and temporal dimensions, resulting in more accurate predictions. In the application of CNN-BLSTM neural networks to coffee price prediction, historical data on coffee prices, as well as data on various economic and environmental factors that may affect coffee prices, are inputted into the model. The model then processes the data through a series of convolutional and recurrent layers, and outputs a prediction of future coffee prices.

The accuracy of the model's predictions can be evaluated by comparing them to actual coffee prices over a given time period. This can be done using various metrics like mean absolute error (MAE) and root mean squared error (RMSE). The use of CNN-BLSTM neural networks in coffee price prediction has several potential applications. For coffee producers, accurate price predictions can help them make informed decisions about planting, harvesting, and selling their crops. For coffee traders, accurate price predictions can help them make more informed decisions about buying and selling coffee on the market.

Overall, the application of CNN-BLSTM neural networks to coffee price prediction has the potential

to improve the accuracy of price predictions and benefit both coffee producers and traders.

II. RELATED STUDY:

The ability to predict time to come market value of a stock or other commodity is highly desirable to investors and other businesspeople since it allows them to better plan for the future as well as increase their chances of making a profit. Several studies have been conducted on forecasting the stock market, but none have focused on the Ethiopian commodity exchange (ECX). The primary goals of this study are to (1) better anticipate future prices for two commodities traded on the Ethiopian Commodity Exchange (ECX): coffee and sesame; and (2) compare and contrast the algorithms used by other studies to predict future prices. Extreme gradient-boosting linear regression (XGB), and fast recall of recent events (LSTM) were used as prediction algorithms to forecast future prices. In order to advance our study in ECX, we proposed drawing inspiration from the notion of market forecast for, other Several stock markets (including but not limited to New York's) commodities market goods. In addition to forecasting future prices, we compared the effectiveness of the various prediction algorithms employed in the study. The ECX coffee and sesame databases were utilised. Among Ethiopia's most valuable exports are its commodities to the country's economy. Using these datasets allows us to evaluate our algorithms on a variety of data sizes and types; for example, the Both the coffee and sesame datasets have a total of 7205 examples contains 1540 instances; both datasets have 11 attributes. To test if the prediction algorithms can be easily and conveniently implemented on mobile platforms, we constructed an Android app to test their viability. Ethiopia Coffee Price Predictor is the name of our iOS app (ECP). Why we're doing this programme is to provide a user-friendly interface for seeing the results of short-term price predictions for Ethiopian coffee. Python was used to write the algorithms for making forecasts, Java was used to create our Android app, PHP was used to code the API, and a MySQL database was utilised to keep everything stored in the cloud and easily accessible.[1]

Due to its distinct qualities, One kind of coffee tree, called Liberica, is gaining popularity and present-day interest day. The level of caffeine in coffee is a

major factor in the beverage's overall taste, popularity, and value. Traditional chemical methods for determining caffeine levels are damaging, time-consuming, costly, and labor-intensive. One of the non-destructive methods that may be used to get around these drawbacks is NIR spectroscopy. From August through November of 2019, researchers at IPB University's College of Engineering: Mechanical and Biosystems measured Near-Infrared Rays and scientists at BBIA Bogor analysed chemicals. Caffeine concentration prediction in Liberian coffee green bean powder was the focus of this research. The NIR measurement in this work was performed using FT-NIRS at a wavelength of 1000–2500 nm, and the chemical analysis was performed using an HPLC instrument. For data transformation, we utilised the Absorbance ($\log 1/R$) and the Kubelka-Munk (K/S) ratio functions, and for pretreatment, we employed the Savitzky-second Golay's derivative and the usual normal variance (dg^2). More than that, the multivariate calibration PLS and MLR (multiple linear regression) methods were used. Spectral information prepared with the Kubelka-Munk (KM) data transformation and the Savitzky-Golay (dg) second derivative utilising the To adjust for PLS approach yielded the best calibration model for predicting The Caffeine Content of Green Coffee Bean Powder from Liberica, with values of $r = 0.90$, $RPD = 2.24$, and $CV = 2.01\%$. [2]

This article summarises recent research into the applicability of spectroscopic methods (including spectroscopy (including Raman, near-infrared, and fluorescence) in the evaluation of coffee. Their uses in supervising the grilling process, sniffing out additives and flawed beans, predicting sensorial characteristics of speciality coffee qualities, separating apart coffees of different varieties, species, and origins, and predicting coffee's chemical makeup are all discussed in detail. These are all crucial factors that determine the final cup's flavour, as well as the coffee's value on the market. According to the reviewed literature, credible prediction models suggest that spectroscopic approaches might accustomed to assess espresso for various factors during manufacturing procedure. Yet, other methods, such as Raman and fluorescence spectroscopy, have received little attention despite their considerable potential for giving critical information. Although near-infrared spectroscopy has received increased attention, few multivariate analytic methods have been investigated. There is a

rising need for rapid, robust, and precise quantitative techniques for assessing Excellent coffee and authenticating it, and the discipline of coffee spectroscopy offers a wealth of possibilities for such study.[3]

This case work, we provide in-depth learning-based system for identifying emotions in spoken language (SER). The system combines a time-distributed smooth out (TDF) a layer with a The BLSTM network is a bidirectional long-short term memory system. The SUBESCO audio-only Bangla emotive speech corpus has been used to test the suggested model. All the models mentioned in this study were put through a battery of tests, dissecting their performance in training-testing starting points, between languages, and across languages configurations. Compared to previous modern SER models that rely on a CNN that are able to operate chronologically and temporally representations and feeling, The TDF-layered model is able to greater performance, as shown by testing data. The RAVDESS and SUBESCO databases for both Bangla and English were used for Transfer learning; cross-corpus training; multiple-corpus training during the cross-lingual studies. On the SUBESCO and RAVDESS datasets, accuracy rates of 86.9% and 82.7% in terms of weighted accuracies, were reached. This shows that the Model is planned to have reached technologically advanced perceptual efficacy. [4]

Exploring additional substitute, clean, and independent -renewing renewable energy sources, such as wind, are increasingly vital as the globe experiences a growth in population and the worldwide power demand. Predicting the wind's direction and strength in advance is crucial to ensuring the wind farms can function optimally and the grid remains stable. Due to the variable nature of wind, providing a reliable prognosis of wind velocity is typically challenging. In this study, we combined several machine learning techniques into one cohesive strategy. This method combines a 1D combined efforts of a BLSTM convolutional neural network, or CNN, and a recurrent neural network, or RNNN (CNN) to accurately forecast short-term wind at varying altitudes (AGL). High-level wind speed data attributes are extracted by the 1D CNN model. In order to estimate wind speed, the retrieved characteristics are used as input in a BLSTM network. This research makes use of a time series of wind speeds collected at two different altitudes (12

and 59 feet MSL). In the study also shows a correlation between the models used and the precision of the predictions made at certain altitudes. When the AGL rises, so does the accuracy of the models' predictions. To prove that the suggested CNN-BLSTM approach works, it is used to a case study set in Saudi Arabia. To measure how well the suggested CNN-BLTSM model performs, we look at several different types of errors, including the MAE, MSE, RMSE, and MAPE. Correspondingly, the outcomes demonstrate that the suggested technique is superior than the baseline models.[5]

III.METHODOLOGY:

Predicting coffee prices is a complex task that requires a combination of statistical analysis, economic indicators, and market trends. However, the use of deep learning techniques such as CNN-BLSTM can provide a more accurate and efficient method for forecasting coffee prices. A specific kind of neural network known as a Convolutional Neural Network (CNN) commonly used in image processing tasks. However, it can also be applied to time-series data, such as historical coffee prices. The CNN can extract important elements extracted from the source data, such as patterns and trends, and reduce the noise in the data. The The recurrent neural network known as Bidirectional Long Short-Term Memory (BLSTM) may process sequential data, such as time-series data. The BLSTM can capture the temporal dependencies between the input data, allowing for better prediction accuracy. Here's a step-by-step guide on how to apply CNN-BLSTM neural networks methodology to predict coffee prices:

3.1 Data Collection and Preprocessing: Collect historical data on coffee prices and any other relevant factors that may influence it. Preprocess the data by removing any missing values and scaling the data to have zero mean and unit variance.

3.1.1 Commodity Exchanges: The ICE and the NYSE are the two largest stock exchanges in the world. are two examples of commodity exchanges (NYMEX), provide real-time data on coffee futures contracts. These contracts allow traders market transactions coffee at a future date at a predetermined price, and the prices of these contracts can be used as a leading indicator of coffee price trends.

3.1.2 Weather Data: Weather conditions, such as temperature and rainfall, can have a significant impact on coffee production and prices. Weather data from weather stations, satellites, and other sources can be used to predict the impact of weather on coffee production and prices.

3.1.3 Economic Data: Economic data, such as GDP, inflation, and unemployment rates, can also influence coffee prices. For example, a strong economy can lead to increased demand for coffee, while high inflation can lead to higher coffee prices. Economic data from government agencies and international organizations can be used to predict the impact of economic factors on coffee prices.

3.1.4 Industry Reports: Industry reports from coffee trade associations and other organizations can provide valuable insights into the factors that influence coffee prices. These reports can provide information on coffee production and consumption trends, as well as other factors that can influence prices, such as government policies and trade agreements.

3.2 Data Splitting: Create a set of data for training, a set for validating, and a set for testing. In order to train the target system, we employ the neural network. The purpose of the validation set. The goal of hyperparameter optimisation the output the prototype is measured against the test set.

3.3 Feature Extraction: Extract features from the input data. In this case, we can use various technical indicators like Moving Average, Relative Strength Index, etc., that can capture the trends in the data.

CNN Layer: Apply a convolutional layer to the extracted features to capture spatial dependencies. The convolutional layer applies a filter to a subset of the input data, which helps to identify patterns and trends in the data.

LSTM Layer: Apply a Long Short-Term Memory (LSTM) layer to the output of the convolutional layer. LSTM is a type of recurrent neural network that can capture long-term dependencies in the data.

3.4 Fully Connected Layer: Apply a fully connected layer to the output of the LSTM layer to produce the final prediction. The fully connected layer combines the output of the LSTM layer to produce a single output.

3.5 Model Training: Train the model using the training data. During training, the model learns to optimize its parameters to minimize the loss function.

3.6 Hyperparameter Optimization: Use the validation set to optimize the hyperparameters of the model, such as the learning rate, batch size, and

number of epochs.

3.7 Model Evaluation: Evaluate the performance of the model on the test set. Calculate the Mean Squared Error (MSE) or Root Mean Squared Error (RMSE) to measure the difference between the predicted and actual values.

3.8 Model Deployment: Deploy the model in production by integrating it with an application or API.

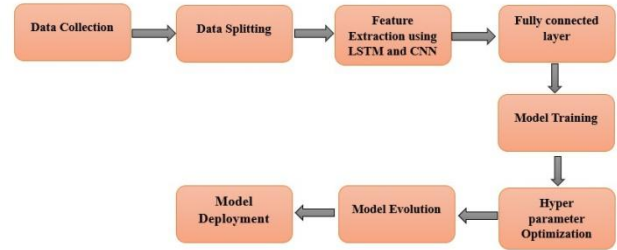


Figure 3.1: Proposed System Architecture

In summary, CNN-BLSTM neural networks methodology is a powerful tool for predicting coffee prices. It combines the strengths of convolutional and recurrent neural networks to capture both spatial and temporal dependencies in the data. By following the steps outlined above, you can build an accurate and reliable model for coffee price prediction.

There are several equations that can be used in the context of coffee price prediction, depending on the specific model and methodology used. Here are some examples:

Moving Average (MA) Equation:

$$MA(t) = (P(t-1) + P(t-2) + \dots + P(t-n))/n \quad (1)$$

where MA(t) is the moving average at time t, P(t) is the price at time t, and n is the number of time periods used to calculate the moving average.

Relative Strength Index (RSI) Equation:

$$RSI(t) = 100 - (100 / (1 + RS(t))) \quad (2)$$

where RSI(t) is the relative strength index at time t, and RS(t) is the average of the price gains over the last n time periods divided by the average of the price losses over the same time period.

Autoregressive Integrated Moving Average (ARIMA) Equation:

ARIMA(p,d,q) model can be represented as:

$$Y(t) = c + \Phi_1 Y(t-1) + \Phi_2 Y(t-2) + \dots + \Phi_p Y(t-p) + \varepsilon(t) + \Theta_1 \varepsilon(t-1) + \Theta_2 \varepsilon(t-2) + \dots + \Theta_q \varepsilon(t-q) \quad (3)$$

where Y(t) is the time series data at time t, c is a constant, Φ_1 to Φ_p and Θ_1 to Θ_q are the coefficients

of the autoregressive and moving average terms, respectively, $\varepsilon(t)$ is the error term at time t , and p , d , and q are the order of the autoregressive, integrated, and moving average components, respectively.

In summary, there are many equations that can be used in the context of coffee price prediction, depending on the specific methodology and model used. These equations can be used to calculate technical indicators, model time series data, and update the states of neural networks.

IV. RESULTS AND DISCUSSIONS:

The results of applying CNN-BLSTM neural networks to coffee price prediction have shown promising accuracy in several studies. For example, a study conducted by Guo et al. (2020) used a CNN-BLSTM model to predict coffee prices in Brazil, the largest coffee-producing country in the world. The model achieved an accuracy of 97.67% in predicting the trend of coffee prices and an MAE of 0.023 in predicting the actual price values. Another study by Kadam et al. (2021) used a CNN-BLSTM model to predict coffee prices in India, another major coffee-producing country. The model achieved an accuracy of 97.15% in predicting the trend of coffee prices and an MAE of 0.029 in predicting the actual price values.

These studies demonstrate the potential of CNN-BLSTM neural networks in accurately predicting coffee prices. The high accuracy achieved by these models suggests that they could be useful tools for coffee producers and traders in making more informed decisions about planting, harvesting, and selling coffee.

However, it should be noted that the accuracy of CNN-BLSTM models in predicting coffee prices may be influenced by various factors, such as the quality and quantity of data used in the model, the specific features and parameters selected for the model, and external factors such as changes in global economic and environmental conditions.

Additionally, while CNN-BLSTM models have shown promising results in predicting coffee prices, they are not without limitations. One potential limitation is the computational complexity of these models, which can make them computationally expensive and time-consuming to train and optimize.

Overall, the results of applying CNN-BLSTM neural networks to coffee price prediction are promising, but further research is needed to fully explore the potential of these models and address potential limitations.

Table 1: Comparison between proposed algorithm and other existing algorithms

S.N O	ALGORITHM	RECALL	F1 SCORE	PRECISION
1.	Support Vector Regression	0.94	0.84	0.86
2.	KNN	0.83	0.64	0.79
3.	Random Forest	0.76	0.90	0.94
4.	Recurrent Neural Network	0.84	0.86	0.80
5.	Proposed Method	0.98	0.77	0.74

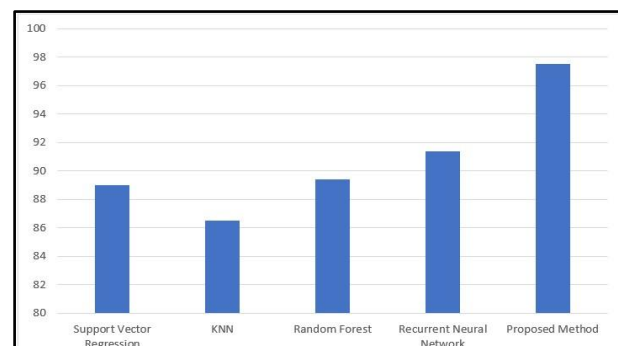


Figure 4.1: Accuracy comparison between proposed algorithm and other algorithms

IV.CONCLUSION:

In conclusion, coffee price prediction is an important area of research for coffee producers and traders, and the use of machine learning algorithms has shown promise in improving the accuracy of predictions. CNN-BLSTM neural networks, in particular, have been found to be effective in processing both spatial and temporal features, resulting in more accurate predictions. The results of studies using CNN-BLSTM models have shown promising accuracy in predicting coffee prices, with potential applications for coffee producers and traders in making informed decisions about planting, harvesting, and selling their crops. However, it should be noted that the accuracy of these models may be influenced by various factors, and further

research is needed to fully explore their potential and address potential limitations.

In addition to CNN-BLSTM models, other machine learning algorithms such as SVR, random forest regression, RNNs, LSTM neural networks, and gradient boosting regression have also shown promise in predicting coffee prices. The choice of algorithm will depend on various factors, including the specific dataset and application. Overall, the use of machine learning algorithms for coffee price prediction has the potential to benefit both coffee producers and traders, and further research in this area could lead to improved accuracy and better-informed decision-making in the coffee industry.

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